THE COMPARISON BETWEEN THE DERIVED CATEGORY OF QUASI-COHERENT SHEAVES AND MODULES OVER THE STRUCTURE SHEAF IS AN EQUIVALENCE

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Allowed standard prerequisites (editorial).

- The adjunction between the comparison functor and the right derived functor of the coherator must have both unit and counit isomorphisms. used at proof lines 1728-1734. Translates the equivalence statement into the two displayed isomorphisms.
- On an affine open, and for pushforwards along affine open immersions, the coherator and the comparison functor are understood. invoked at proof lines 1760-1798. Supplies the two statements used in the induction step.
- The support of an object is a closed subset, and a closed subset contained in a union of affine opens supports the induction. invoked at proof lines 1735-1745. Defines the hypothesis on which the induction runs.

Source reference. Original definition 08DA: sources/stacks-perfect/pinned-original-source.tex, source lines 1005–1005. The induction hypothesis is phrased in terms of this definition; it fixes notation, and every use of the notion is inside the counted span.

Source reference. Original lemma 014A: sources/stacks-chow/tags.txt, source lines 1–1. Invoked in the induction step at proof lines 1770-1780 to reduce the required isomorphism to the two cases; the reduction itself is inside the counted span.

Source reference. Original lemma 08D9: sources/stacks-perfect/pinned-original-source.tex, source lines 1670–1670. Cited at proof lines 1786-1793 for the affine case of the induction step; the induction argument that consumes it is inside the counted span.

Source reference. Original lemma 08D8: sources/stacks-perfect/pinned-original-source.tex, source lines 1609–1609. Cited together with the previous label at proof lines 1793-1798 to finish the induction step; the final conclusion is inside the counted span.

Source reference. Original equation 06VT: sources/stacks-perfect/pinned-original-source.tex, source lines 289–289. The statement refers to the comparison functor

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by this label; it is a pointer to the display, and the proof works with that functor inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Suppose that for every affine open $U \subset X$ the right derived functor*

$$\Phi : D(QCoh(\mathcal{O}_U)) \to D(QCoh(\mathcal{O}_X))$$

of the left exact functor $j_* : QCoh(\mathcal{O}_U) \to QCoh(\mathcal{O}_X)$ fits into a commutative diagram

$$\begin{array}{ccc} D(QCoh(\mathcal{O}_U)) & \xrightarrow[i_U]{} & D_{QCoh}(\mathcal{O}_U) \\ \Phi \downarrow & & \downarrow Rj_* \\ D(QCoh(\mathcal{O}_X)) & \xrightarrow{i_X} & D_{QCoh}(\mathcal{O}_X) \end{array}$$

Then the functor (06VT)

$$D(QCoh(\mathcal{O}_X)) \longrightarrow D_{QCoh}(\mathcal{O}_X)$$

is an equivalence with quasi-inverse given by RQ_X.

Proof. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$ and let A be an object of $D(QCoh(\mathcal{O}_X))$. We have to show that the adjunction maps

$$A \to RQ_X(i_X(A)) \quad \text{and} \quad i_X(RQ_X(E)) \to E$$

are isomorphisms. Consider the hypothesis H_n: the adjunction maps above are isomorphisms whenever E and $i_X(A)$ are supported (Definition 08DA) on a closed subset of X which is contained in the union of n affine opens of X. We will prove H_n by induction on n.

Base case: $n = 0$. In this case $E = 0$, hence the map $i_X(RQ_X(E)) \to E$ is an isomorphism. Similarly $i_X(A) = 0$. Thus the cohomology sheaves of $i_X(A)$ are zero. Since the inclusion functor $QCoh(\mathcal{O}_X) \to Mod(\mathcal{O}_X)$ is fully faithful and exact, we conclude that the cohomology objects of A are zero, i.e., $A = 0$ and $A \to RQ_X(i_X(A))$ is an isomorphism as well.

Induction step. Suppose that E and $i_X(A)$ are supported on a closed subset T of X contained in $U_1 \cup \ldots \cup U_n$ with $U_i \subset X$ affine open. Set $U = U_n$. Consider the distinguished triangles

$$A \to \Phi(A|_U) \to A' \to A[1] \quad \text{and} \quad E \to Rj_*(E|_U) \to E' \to E[1]$$

where Φ is as in the statement of the lemma. Note that $E \to Rj_*(E|_U)$ is a quasi-isomorphism over $U = U_n$. Since $i_X \circ \Phi = Rj_* \circ i_U$ by assumption and since $i_X(A)|_U = i_U(A|_U)$ we see that $i_X(A) \to i_X(\Phi(A|_U))$ is a quasi-isomorphism over U. Hence $i_X(A')$ and E' are supported on the closed subset $T \setminus U$ of X which is contained in $U_1 \cup \ldots \cup U_{n-1}$. By induction hypothesis the statement is true for A' and E'. By Derived Categories, Lemma 014A it suffices to prove the maps

$$\Phi(A|_U) \to RQ_X(i_X(\Phi(A|_U))) \quad \text{and} \quad i_X(RQ_X(Rj_*E|_U)) \to Rj_*(E|_U)$$

are isomorphisms. By assumption and by Lemma 08D8 (the inclusion morphism $j : U \to X$ is flat, quasi-compact, and quasi-separated) we have

$$RQ_X(i_X(\Phi(A|_U))) = RQ_X(Rj_*(i_U(A|_U))) = \Phi(RQ_U(i_U(A|_U)))$$

and

$$i_X(RQ_X(Rj_*(E|_U))) = i_X(\Phi(RQ_U(E|_U))) = Rj_*(i_U(RQ_U(E|_U)))$$

Finally, the maps

$$A|_U \to RQ_U(i_U(A|_U)) \quad \text{and} \quad i_U(RQ_U(E|_U)) \to E|_U$$

are isomorphisms by Lemma 08D9. The result follows. □